

Wireless Attacks

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Wireless Attacks

Objectives:

- 4.7: Given a scenario, perform wireless attacks using the appropriate tools.
- 4.8: Given a scenario, perform wireless attacks.

Video Overview

In this video, we'll delve into the foundational aspects of wireless attacks, exploring how these techniques are used to disrupt and exploit wireless networks and devices. Understanding these attacks is useful for aspiring cybersecurity professionals as it directly relates to real-world scenarios you might encounter in penetration testing and defense. We will cover various attack categories, including Wi-Fi, Bluetooth, RFID, and NFC, providing you with a comprehensive overview of the vulnerabilities inherent in common wireless technologies. This knowledge will not only prepare you for the PenTest+ exam but also equip you with practical insight.

Key Topics Covered

- **Wireless Attack Categories:** Jamming (Denial of Service), spoofing, and data corruption/manipulation.
- **Wi-Fi Attack Tools:** Aircrack-ng suite (airmon-ng, airodump-ng, aireplay-ng), Kismet, WPAD, WiFi-Pumpkin.
- **Deauthentication Attacks:** Forcing client disconnection from Wi-Fi networks using aireplay-ng.
- **Handshake Capture and Cracking:** Obtaining encryption information (WEP, WPA, WPA2, WPA3) via deauthentication and cracking with aircrack-ng.
- **Wi-Fi Protected Setup (WPS) Vulnerabilities:** Exploiting short PINs and push-button access for network entry.
- **Evil Twin Attacks:** Setting up rogue access points to intercept user data.
- **Bluetooth Attacks:** Bluejacking (sending unsolicited messages), Bluesnarfing (unauthorized data access), Bluetooth spamming, and Bluetooth Low Energy (BLE) attack vulnerabilities.
- **RFID and NFC Attacks:** Cloning technologies (e.g., with Proxmark) and exploiting lack of encryption for unauthorized access or payments.

Student Notes Section